

Designing and evaluating a system for studying EarEEG signals

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ABSTRACT

Electroencephalography (EEG) allows the study of the brain in humans with applications in areas of Psychology, Brain-Computer Interfaces, and Neuromarketing. The conventional systems limit the applications of EEG to lab environments, but recent developments towards portable EEG have allowed brain studies outside the laboratory in more realistic situations. One such direction is the Ear EEG, a wearable concept for recording the EEG using an ear worn device. This paper discusses a system design to study the Ear EEG compared to the scalp EEG using the OpenBCI. We evaluate the signal quality obtained with Ear EEG using a customized earpiece and demonstrate the feasibility of recording alpha attenuation using Ear EEG.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; *Mobile devices*.

KEYWORDS

EEG; EarEEG

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1 INTRODUCTION

There exist several brain imaging modalities, such as Electroencephalography (EEG), Magnetoencephalography (MEG), Functional near-infrared spectroscopy (fNIRS), and functional Magnetic resonance imaging (fMRI), used to understand the anatomy and function of the brain for monitoring and diagnostic purposes. These modalities have their advantages and disadvantages in terms of spatial and temporal resolution, portability, and cost, and most are not well-suited for mobile and lifestyle integration. EEG is widely used

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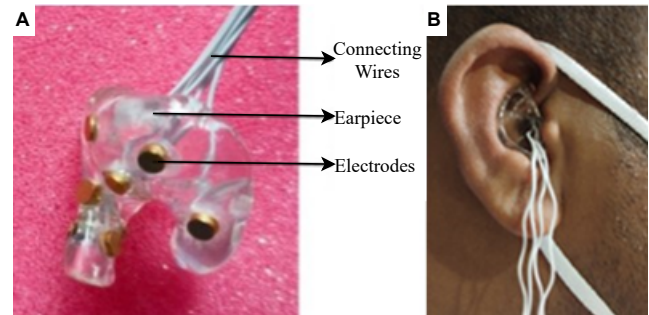


Figure 1: (A) Custom designed ear mould with electrodes placed and (B) A subject wearing the ear mould

to investigate the brain in applications such as neurophysiology, brain-computer interfaces, and neuromarketing.

EEG is a non-invasive recording of the brain's electrical activity, typically by placing electrodes on the scalp's surface. The EEG on the scalp estimates synaptic action at large scales closely related to behavior and cognition. Thus, EEG can provide a genuine window on the mind [10]. The clinical systems to record EEG require a conductive gel applied to the electrode sites. The use of conductive gels causes discomfort to the subjects and limits the recording time as the signal quality degrades with the drying of the gels. Although EEG analysis has advanced in academic and professional contexts, the complexity of clinical EEG systems and the requirement of specially trained personnel have made them unsuitable for mobile and lifestyle integration.

Researchers in recent years are working towards developing portable EEG technologies, offering a wireless, compact, and hassle-free EEG system to enable monitoring of the brain to understand the neural correlates of behavior, mental processes, and sleep. [12]. Wearable EEG is a technology that enables the translation of neuroscience from the laboratory to the real-life environment. The Ear EEG is a wearable concept of recording the EEG in the ear introduced by Looney et al. [8] demonstrating the feasibility of recording the ear EEG. The Ear EEG has shown to produce signal-to-noise ratios comparable to those from conventional EEG electrode placements and is robust to familiar sources of artifacts. The studies have been conducted on ear EEG to check its effectiveness in recording auditory and visual evoked potentials[7].

In this paper, we have presented the early results of the feasibility of Ear EEG and the methodology to build a rigorous and comprehensive system using OpenBCI to study a larger population of subjects and to span several protocols, and in this way, assess the use cases of Ear EEG compared to scalp EEG.

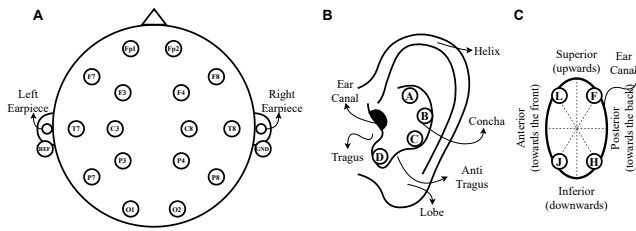


Figure 2: Electrode Configuration on Scalp and Ear. (A) Electrode placement on the scalp according to the 10-20 system. The reference electrode is placed on the left earlobe, and the ground electrode is placed on the right earlobe (B) Sketch of the exterior part of the left ear, showing the four regions corresponding to the electrode labels A through D. (C) Cross-sectional view of the left ear canal (sagittal plane) showing the electrode labels in the left ear canal. The electrodes are labeled based on the direction relative to the vertical axis and not based on the depth in the ear canal

2 SYSTEM DESIGN

2.1 Earpiece

This section discusses the design of customized earpieces. The earpieces were developed according to the anatomical shape of the subjects. An audiologist captured the impression of the ear by cleaning the ear and inserting a cotton ball with a thread inside the ear canal. Then, silicone was filled into the ear canal and the concha region. The impression was then carefully removed once the silicone had dried. The ear impression was then used to create a 3D ear mould. Then electrodes were placed according to the positions described in Figure 2. Various electrode materials were simulated [2], and gold-plated electrodes were selected. Gold-plated surface mount PCB contact was used as ear-EEG electrodes. The electrode had a diameter of 3mm and was 1.9mm thick. Figure 1 shows the earpiece design with electrodes mounted and a subject wearing the earpiece.

2.2 Electrode Configuration

Scalp. The ScalpEEG consisted of 16 electrodes according to the international 10-20 system and were FP1, FP2, F3, F4, F7, F8, C3, C4, T7, T8, P3, P4, P7, P8, O1, O2. The scalp electrode positions are shown in Figure 2A. Dry Comb electrodes were used for scalpEEG.

Ear. The EarEEG consisted of eight electrodes in each ear, with four electrodes placed in the ear canal and concha. The electrodes labeling convention follows [6]. The electrodes are labeled Exy, where x denotes the left(L) or right(R) ear, and y denotes the position in the ear. Positions A through D covers the concha region at four different locations. Positions F, H, J, and L, denote the ear canal electrodes. Figure 2B and 2C depicts the positions of the electrodes in the ear concha and canal.

2.3 Hardware

The OpenBCI Cyton-Daisy Biosensing Boards were used for recording EEG. The OpenBCI system is a low-cost open-source EEG amplifier with a proven record as an alternative to medical-grade EEG

amplifiers [4] [11]. The OpenBCI Cyton is an eight-channel system with a sampling rate of 250Hz over Bluetooth, while 16 channels can be used with the add-on Daisy Board at a reduced sampling rate of 125Hz. The system uses the ADS1299, a 24-bit biopotential measurement integrated circuit developed by Texas Instruments. A PIC32-based microcontroller is implemented on the board, providing plenty of local memory and fast processing speeds. In addition, A 3-axis accelerometer (LIS3DH). A module for storing data in a micro SD card. The application to communicate with OpenBCI is an open-source application written with Processing language. The amplifier supports both wet and dry electrode configurations and has support for external sensors, and was chosen given the capability for customizing the recording configurations. There also exists a few issues with the OpenBCI, the main concerns being low sampling rate and high sensitivity of electrical noise to resolve which additional hardware is required.

3 METHODOLOGY

3.1 Experiment Design

Protocols. Five different protocols and the alpha attenuation task were chosen as stimuli for the subjects. The motivation for choosing these stimuli was to activate the five broad lobes and to study their influence on Ear EEG

Alpha Attenuation Task The subjects were instructed alternately to rest with open and closed eyes for 1 minute each. Auditory cues were presented at one-minute intervals, instructing the subjects to alternate between performing eyes-open and relaxing with eyes closed. This activity of opening and closing eyes is known to attenuate the alpha band (8-13Hz) in EEG.

Four Word Sentence Task The processing of words and sentences and memory tasks have been associated with the parietal lobe. The subject is presented with a set of four-letter words, following which the subject is to generate a meaningful sentence with words starting from each letter of the word [1].

Go-No-Go Task The role of the frontal cortex in response inhibition. The frontal cortex has been reported to be active during the infrequent stimulus. The Go/No-go task is an experimental protocol used to determine the response inhibition of the subject. The subject must respond by pressing a button when they see a “go” condition and not respond when they see the “no-go” condition.

Steady State Visual Evoked Potential (SSVEP) The human visual cortex can perceive light modulation to frequencies up to 75Hz, which is reported to be most sensitive around 10Hz. The SSVEP is typically induced by flashing light through a monitor or using a light source such as LED [7].

Motor Task Motor task can be defined as the subject’s mental process of motor activity with actual motor movement. It is a well-reported fact that motor movement causes event-related synchronization (ERS) or event-related desynchronization (ERD) in the central μ (8-12Hz) and β (13 - 35Hz) bands [9].

Auditory Steady State Response (ASSR) Auditory steady-state responses (ASSR) were reported as a method to assess the hearing threshold level objectively. Amplitude-modulated narrow or broad-band sound signals can evoke the ASSRs. The neural encoding of the modulating frequency is visible in the frequency spectrum of the EEG [7].

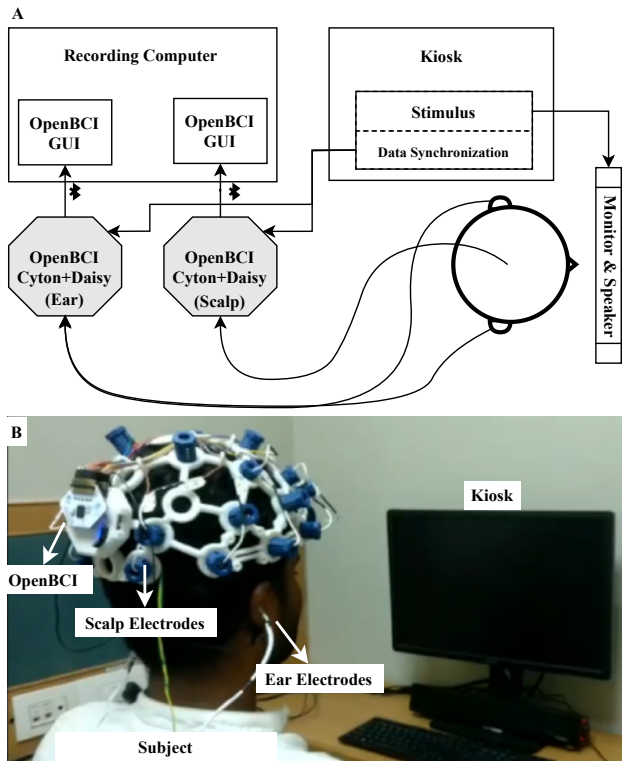


Figure 3: A) The block diagram of the recording setup, B) A subject in the recording setup wearing the scalp and ear electrodes

Kiosk. The protocols have been deployed on a kiosk developed in-house[5]. The kiosk has been developed predominantly in the Java platform and extensively tested on Microsoft Windows. Custom libraries were written to handle the stimulus drawing, hardware interfacing, data handling, etc. The kiosk is a subject-friendly tool and does not need the experimenter's intervention during the task to give the subject's privacy during the experiments.

3.2 Data Acquisition

The data was collected from 6 subjects after informed consent. For this study only the alpha attenuation paradigm is considered. The EEG recording was done using two OpenBCI Cyton-Daisy amplifier systems, each for the scalp and ear. The raw data was recorded using the OpenBCI GUI at a sampling rate of 125Hz. The electrode sites were cleaned using Isopropyl alcohol prior to electrode placement. The Ultracortex Mark iv headset was used to place the scalp electrodes and earpieces for the ear electrodes. The ground electrode of both the scalp and ear was placed on the right earlobe and the reference electrode on the left earlobe. Figure 3A shows the block diagram, and Figure 3B shows a subject in the recording setup.

Data Validation To validate the correctness of the EEG collected few checks were done visually on the OpenBCI GUI. Each channel data was visually inspected to check the signal by instructing the participants to blink their eyes and clench their jaws. Impedance was measured, and made sure low impedance was maintained for

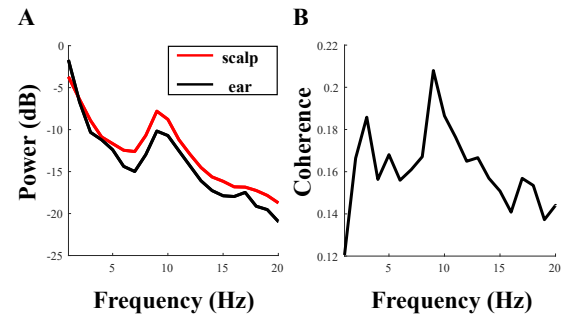


Figure 4: A) Average PSD results of the the scalp and ear electrodes for a Subject for the Baseline recordings, B) Average spectral coherence between the scalp and ear electrodes

good EEG signals. The railed information in the OpenBCI GUI was checked, and made sure that all channels were not railed.

Data Synchronization The data synchronization was at two levels i) to synchronize the scalp and ear EEG and ii) to synchronize the stimulus with EEG data. An Arduino-based system was developed to synchronize the OpenBCI systems with the stimulus. The kiosk communicated with the Arduino over serial communication, and the Arduino responded to these messages by toggling the digital output pin. The same digital pin output was fed to the OpenBCI systems through optical isolators, thus synchronizing the datastreams of the EEG with the stimulus. The event markers were recorded on the OpenBCI using the digital I/O pins in the AnalogRead mode.

3.3 Data Processing

The first step of the data processing was the synchronization of the scalp and ear datasets. The auxiliary analog channel recorded on the two OpenBCIs was used to calculate the event markers and these event markers was then used to align the data streams. Few datasets were excluded from analysis due to noisy recordings. The data preprocessing steps for scalp and ear EEG remained the same.

The data processing was done using EEGLAB, and the steps are as follows. The data was cropped to reject the first and last 10 seconds of data. The data was then highpass filtered to 1Hz. Additionally, the zapline toolkit[3] was used to remove the first and second harmonics of the line frequency (50Hz). Post filtering, manual inspection of the data was done to annotate bad data segments, and the bad segments were rejected. Bad channels were detected using the OpenBCI railed/not railed logic, and the channels were also rejected based on visual inspection.

4 EXPERIMENTAL RESULTS

Quality of Ear EEG. The quality of the Ear EEG was verified by comparing the power spectral density of the ear and scalp EEG. The power spectral density was computed with the multitaper method using a single taper. Figure 4A shows the average power spectral density of a subject's ear and scalp EEG. It was observed that the ear EEG showcased less power compared to the scalp. Spectral coherence was computed to determine the similarity between the scalp and ear EEG across frequencies. The spectral coherence ranges from 0-1, with values close to 1 depicting higher coherence at those frequencies. The spectral coherence between the scalp and ear EEG

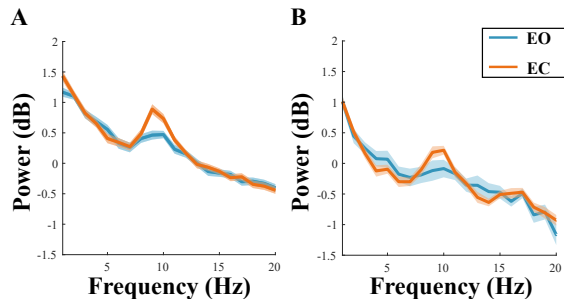


Figure 5: A comparison between eyes open (EO) and eyes close (EC) condition during baseline recordings in A) scalp electrode (O1) and B) ear electrode (ERL)

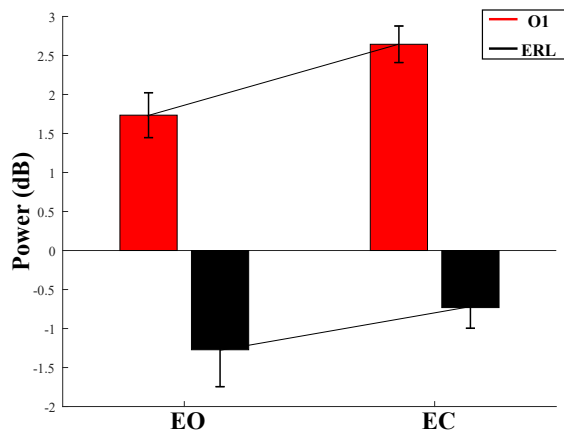


Figure 6: Bar plots depicting the change in total alpha power between eyes open (EO) and eyes close (EC) conditions in scalp electrode (O1) and ear electrode (ERL)

was the maximum around the alpha band, with averages around 0.2. Figure 4B shows the average spectral coherence between the scalp and ear electrodes.

Alpha Attenuation. The data was segmented based on the eyes open and eyes closed markers. The power spectral density was computed, and the absolute power in the alpha band was computed. The results of alpha attenuation for a typical subject (SUB003) are discussed here. Figure 5A for the scalp and Figure 5B for the ear shows the PSD plots for the respective EEG during the Eyes Open (EO) and the Eyes Close (EC) conditions. The mean alpha power of the Ear EEG was found to be lesser by half compared to scalp EEG. The change in power in alpha attenuation was computed as the ratio of absolute power in the alpha band during EC and EO. The Table1 tabulates the change in alpha power shown in Figure 6. The alpha power in the scalp EEG showed a change of 152%, whereas the Ear EEG showed a change of 57% between the EO and the EC conditions.

5 CONCLUSION AND DISCUSSIONS

In this paper, we discussed the system design to study Ear EEG. The methodology to design earpieces and interface them with OpenBCI was described. The scalp and ear EEG were recorded simultaneously

EEG Location	Scalp	Ear
Average Eyes Open alpha power (dB)	1.73	-1.27
Average Eyes Close alpha power (dB)	2.64	-0.73
Change in alpha power	1.52	0.57

Table 1: Alpha power in Scalp and Ear EEG for a subject

with common reference and ground. The subjects performed the alpha attenuation task of EO and EC.

It was observed that Ear EEG is effectively identical to similarly referenced conventional scalp EEG. The Ear EEG measurements had lower power in comparison with scalp EEG. The alpha activity during Eyes Open and Eyes Close in Ear EEG was found to have less discrimination but was completely acceptable. This demonstrates that the bio-potential acquired by the Ear EEG platform is similar to that of conventional scalp electrodes.

This motivates us to study the effect and correlation of the endogenous and exogenous cortical potentials generated in the different lobes due to various protocols, frequency bands, and frequency response characterization on the Ear EEG. Establishing reliable Ear EEG may lead the research toward sleep studies and brain-computer interfaces, which demands stable electrode performance, comfort, and scalable form factor. The aim is to develop an Ear EEG wearable with comfort levels comparable to the modern-day state-of-the-art wireless earbuds.

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